

Name: _____

Date: _____

Teacher: _____

Year 9 2019

Pythagoras and Similar Triangles Test

/ 40



Baldivis
Secondary College

Full working out MUST be shown to get full marks for each question.

Total Time: 55 minutes

Weighting: 9%

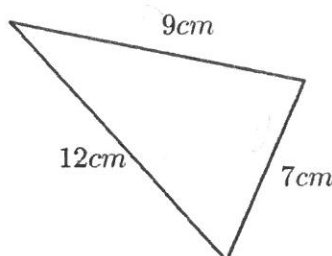
Equipment: To be provided by the student: Pen, pencil, ruler, scientific calculator, 1 - 2-sided page of notes

Question 1

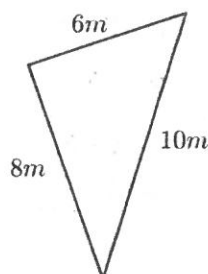
4 marks

Are the following triangles right angled? Justify your answer

a)



b)



$$\frac{12}{10} = 1.2$$

$$\frac{9}{8} = 1.125$$

$$\frac{7}{6} = 1.16$$

No

$$7^2 + 9^2 \neq 12^2$$

$\therefore$ Not Right Angled

No - Sides are not in the same ratio

Yes

$$6^2 + 8^2 = 100$$

$$10^2 = 100$$

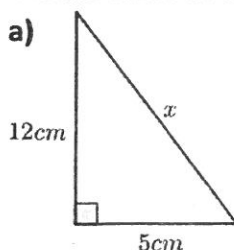
$\therefore$ Pytha Yes

Question 2

9 marks

Find the value of the missing side for the following triangle.

a)



$$x^2 = 12^2 + 5^2 \quad \checkmark$$

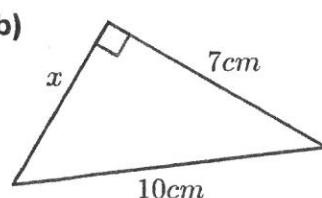
$$= 144 + 25 \quad \checkmark$$

$$= 169 \quad \checkmark$$

$$x = \sqrt{169}$$

$$= 13\text{cm} \quad \checkmark$$

b)



$$x^2 = 10^2 - 7^2 \quad \checkmark$$

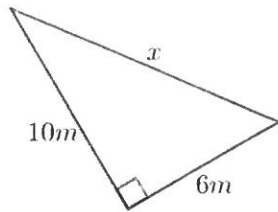
$$= 100 - 49 \quad \checkmark$$

$$= 51 \quad \checkmark$$

$$x = \sqrt{51}$$

$$\approx 7.14\text{cm} \quad \checkmark$$

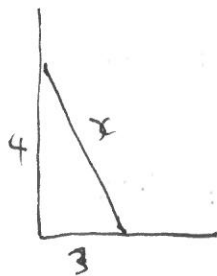
c)



$$\begin{aligned}
 x^2 &= 10^2 + 6^2 \quad \checkmark \\
 &= 100 + 36 \\
 x &= \sqrt{136} \quad \checkmark \\
 &= 11.66\text{m} \quad \checkmark
 \end{aligned}$$

Question 3**3 marks**

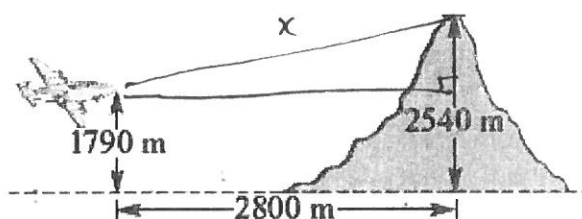
A ladder is leaning against the wall. The top of the ladder is 4 meters high on the wall, and the bottom of the ladder is 3 meters from the wall. Draw a diagram for this and find the length of the ladder.



$$\begin{aligned}
 x^2 &= 3^2 + 4^2 \quad \checkmark \\
 &= 9 + 16 \\
 &= 25 \\
 x &= \sqrt{25} \quad \checkmark \\
 &= 5 \quad \checkmark
 \end{aligned}$$

Question 4**4 marks**

A pilot flying a plane at an altitude of 1790 m sees a mountain peak 2540 m high, 2800 m away from him in the horizontal direction. Find the direct distance between the plane and the mountain peak, correct to the nearest metre.



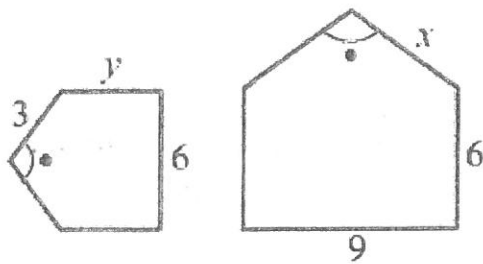
$$\begin{aligned}
 x^2 &= 2800^2 + (2540 - 1790)^2 \quad \checkmark \\
 &\quad + 750^2 \\
 &= 8\,402\,500
 \end{aligned}$$

$$\begin{aligned}
 x &= 2898.71\text{ m} \quad \checkmark \\
 &= 2899\text{ m (nearest metre)} \quad \checkmark
 \end{aligned}$$

Question 5

5 marks

This pair of figures are similar.



- a) Find a scale factor.

$$S.f. = \frac{9}{6} = 1.5 \quad \checkmark$$

- b) Find the value of x .

$$x = 3 \times 1.5 \\ = 4.5 \quad \checkmark$$

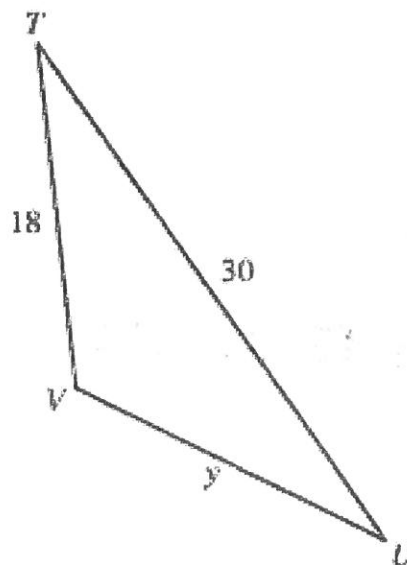
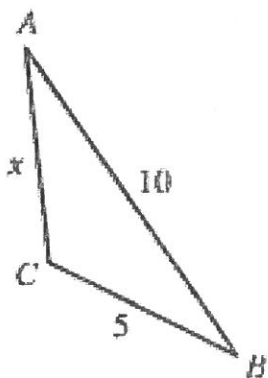
- c) Find the value of y .

$$y = 6 \div 1.5 \\ = 4 \quad \checkmark$$

Question 6

2 marks

Find the unknown side for the following similar triangles:



$$S.f. = \frac{30}{10} = 3 \quad \checkmark$$

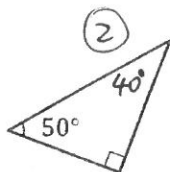
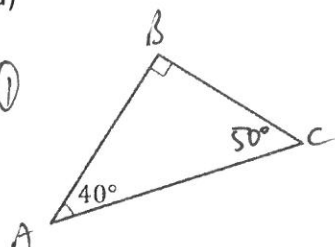
$$x = 18 \div 3 \\ = 6 \quad \checkmark \quad (1)$$

$$y = 5 \times 3 \\ = 15 \quad \checkmark \quad (1)$$

Determine if the following pairs of triangles are similar, and state the similarity test that proves it.

a)

①



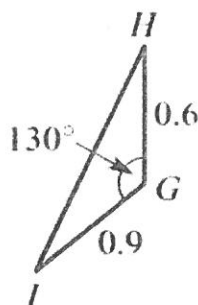
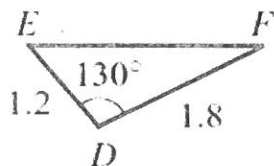
In $\Delta 1$ the missing angle is 50°

In $\Delta 2$ the missing angle is 40° ✓

$\therefore$ Corresponding angles are equal
So the triangles are similar (AAA) ✓

Yes ✓

b)



$$\angle EDF = \angle GHI \quad (130^\circ)$$

$$\frac{ED}{GI} = 2$$

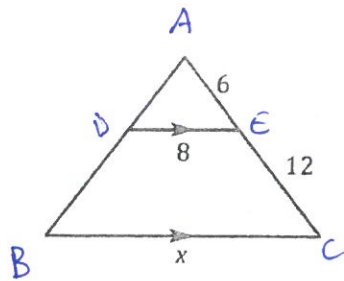
$$\frac{DF}{HI} = 2$$

$$\therefore \Delta EDF \cong \Delta GHI \quad (\text{SAS})$$

Yes

Question 8

4 marks



- a) Give a reason why the two triangles are similar.

$\angle A$ is common to both triangles
 $\angle ADE = \angle ABC$ (corresponding $\angle$'s, $DE \parallel BC$)
 $\angle AED = \angle ACB$ (" " " ")
 $\therefore \triangle ADE \cong \triangle ABC$ (AAA)

- b) Find the value of x .

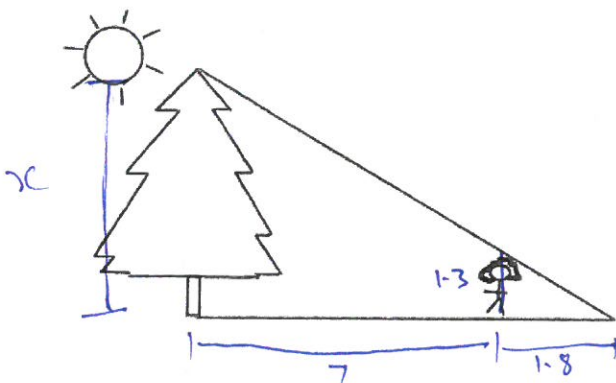
$$\frac{x}{8} = \frac{18}{6} = 3$$

$$x = 3 \times 8 = 24$$

Question 8

3 marks

Tonya is 1.3 meters tall. She stands 7 meters in front of a tree and casts a shadow 1.8 meters long. How tall is the tree?



$$\frac{x}{8.8} = \frac{1.3}{1.8}$$

$$x = \frac{1.3 \times 8.8}{1.8}$$

$$= 6.36 \text{ m}$$

